

Ch-7

Everything is made up of constituent particles. Constituent particles are basic units that make up a larger material or substance.

Interparticle space is the space b/w molecule or atoms of a substance. Eg: When you mix sugar with water, it looks like it vanished but you can still feel its presence in water by tasting. So, conclusion is that during mixing sugar breaks down into its constituent particles & occupies the inter-particle space of water.

Diagram:



Interparticle attraction is a major factor for deciding the state of matter. This force holds the particles & even a slight change in the distance of particles & can weaken these force drastically.

Solids

They have a very strong interparticle force.

Particles are tightly & closely packed

They've a fixed shape & volume.

Particles can only vibrate.

Liquids

The interparticle force is not that strong like solids, but it is present & working.

Particles are not that closely packed, but they are still loosely packed.

They don't have a fixed shape but have a fixed volume.

Particles can move freely but interparticle attraction still keeps them packed in some way.

Gases

The interparticle force in ~~gases~~ gas is negligible. Particles are far apart.

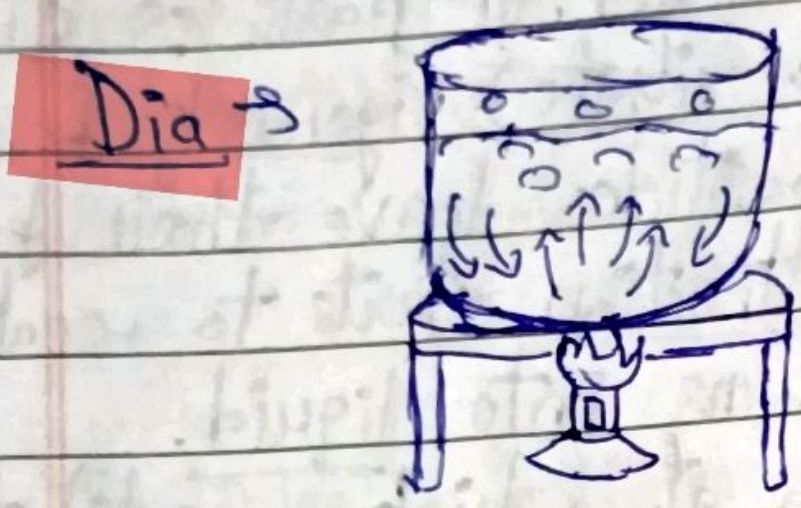
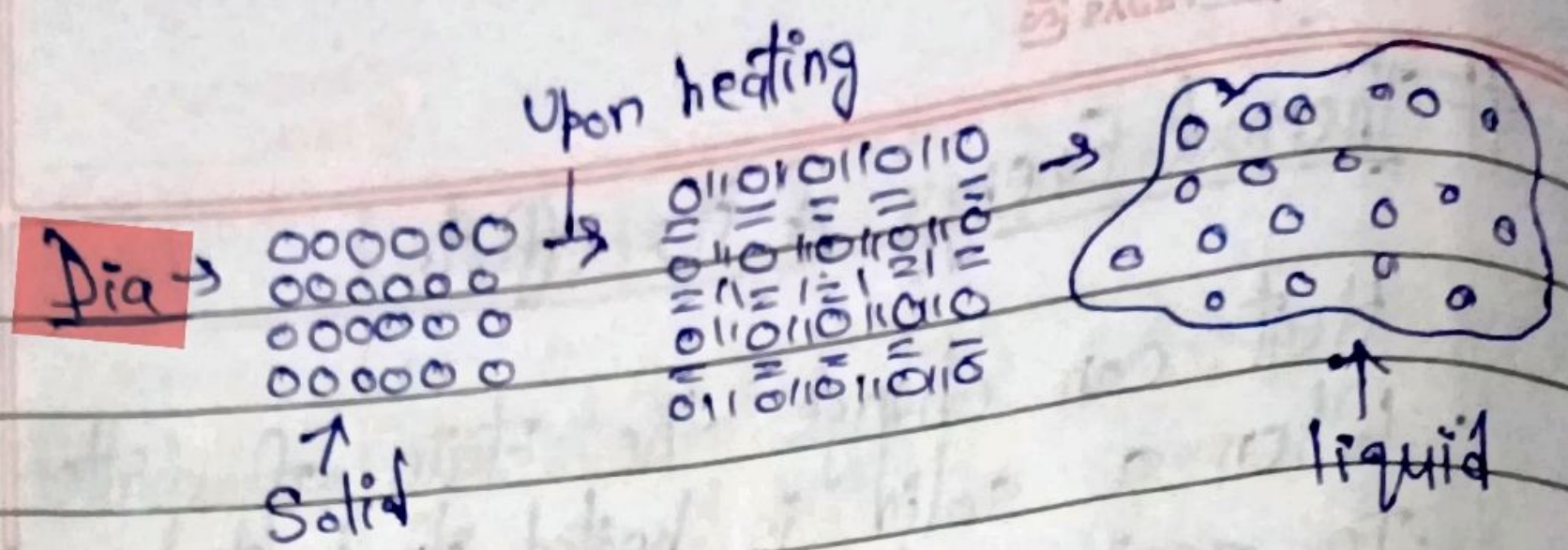
Particles are far apart & are not packed in any way.

They neither have a fixed shape nor a fixed volume.

Particles are not packed in any manner & they can move freely.

Thermal Energy & its effects

- Heat can change the state of matter
- When a solid is heated, its particles start to ~~osc~~ oscillate (or vibrate) vigorously & at a certain stage, particles leave their position & interparticle attraction starts to weaken. Soon, the solid turns into liquid.
- The ~~stage~~ temp. at which a solid turns into a liquid is known as Melting point.
- Melting point depends on the strength of interparticle attraction. Eg: Ice - 0°C
- The temp. at which a liquid turns into vapour at atmospheric pressure is known as Boiling point. Eg - Water: 100°C
- Vapourisation of any liquid also happens ~~at~~ below the boiling point, but it is not that huge & does not occur with bubble formation. This is called Evaporation.
- Boiling occurs with bubble formation & rapid movement at the surface of liquid.



Melting point of Iron is 1538°C & Urea is 133°C

Both liquids & gases can ~~move freely~~ flow & don't have a definite shape. They are classified as fluids.

The interparticle space is filled with literal nothing.

SPM (Suspended particulate matter) refers to tiny dust particles suspended in the air. It is used to tell about Air Pollution.

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PAGE: _____
If you put a chunk of Potassium Permanganate in water, after some time, the water will turn magenta. Why?

⇒ Because water particles are constantly in motion. Due to this constant motion, water pulls the particles of potassium permanganate & later hit them. This causes the change.

If the obj has a very strong interparticle attraction, then water might not be able to break it down & dissolve it. Eg - Sand.

This constant motion is seen in air & other fluids too.

Acharya Kanada

He was an ancient Indian philosopher, who wrote a book named Vaisheshika Sutra, in which he talked about the idea of Parmanu (atom). He considered parmanus, a eternal indivisible particles.